Image Analysis Report

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Executive Summary

The valve and flange assembly exhibit severe atmospheric corrosion with significant metal loss and scaling, indicating a high level of degradation. This level of severity suggests that structural integrity may be compromised and immediate corrective action is required to prevent containment failure.

Key Findings

#	Finding
1	Severe general atmospheric corrosion with thick oxide scaling (rust) across the entire valve body and flange surfaces.
2	Extensive corrosion on flange bolts and nuts, likely resulting in significant cross-sectional area loss and potential for failure under pressure.
3	Complete breakdown of the protective coating system on the piping, valve body, and flange components.
4	Accumulation of moisture and debris around the base due to direct contact with gravel, facilitating accelerated corrosion at the ground interface.
5	Evidence of deep pitting on the flange edges and valve neck, indicating localized metal loss.
6	Internal corrosion state and remaining wall thickness cannot be determined from this external visual inspection.
7	Possible crevice corrosion at the flange-to-gasket interface, which could lead to sudden seal failure.

Recommendations

#	Recommendation
1	Perform immediate Ultrasonic Thickness (UT) gauging on the valve body and adjacent piping to determine if remaining wall thickness meets API 570/ASME B31.3 minimum requirements.
2	Abrasive blast clean the assembly to Sa 2.5 (Near-White Metal) to fully assess the extent of pitting and surface degradation.
3	Replace all flange bolting and nuts with new B7 or equivalent high-strength fasteners, ensuring the use of anti-seize or protective caps.
4	Disassemble the flange connection to inspect gasket seating surfaces for 'wire-drawing' or pitting that would prevent a proper seal.
5	Apply a high-performance, multi-coat industrial epoxy or zinc-rich primer coating system suitable for high-moisture environments.
6	Install proper pipe supports or concrete sleepers to elevate the equipment at least 150mm off the ground to prevent moisture entrapment.
7	If UT results show wall loss exceeding 20% of nominal thickness, prioritize the replacement of the entire valve and flange section.

8	Perform a pressure test (Hydrostatic or Pneumatic) after any repairs or bolt replacements to ensure leak-free integrity before returning to service.